In [ ]:

```
                                                        Pratical No : 03

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Batch : A1
Problem Statement : Implement Gradient Descent Algorithm to find the local minima
function.
```

In [10]:

```
import numpy as np
import matplotlib.pyplot as plt
```

In [11]:

```
def f(x):
    return x**2
```

In [12]:

```
def df(x):
    return 2*x
```

In [15]:

```
def gradient_descent(starting_point, learning_rate, iterations):
    x = starting_point
    history = [x]

    for i in range(iterations):
        grad = df(x)
        x = x - learning_rate * grad
        history.append(x)

    return x, history
```

In [16]:

```
starting_point = 10
learning_rate = 0.1
iterations = 50
```

In [17]:

```
minimum, history = gradient_descent(starting_point, learning_rate, iterations)
```

In [18]:

```
print("Local minimum occurs at x =", minimum)
```

```
Local minimum occurs at x = 0.00014272476927059603
```

In [19]:

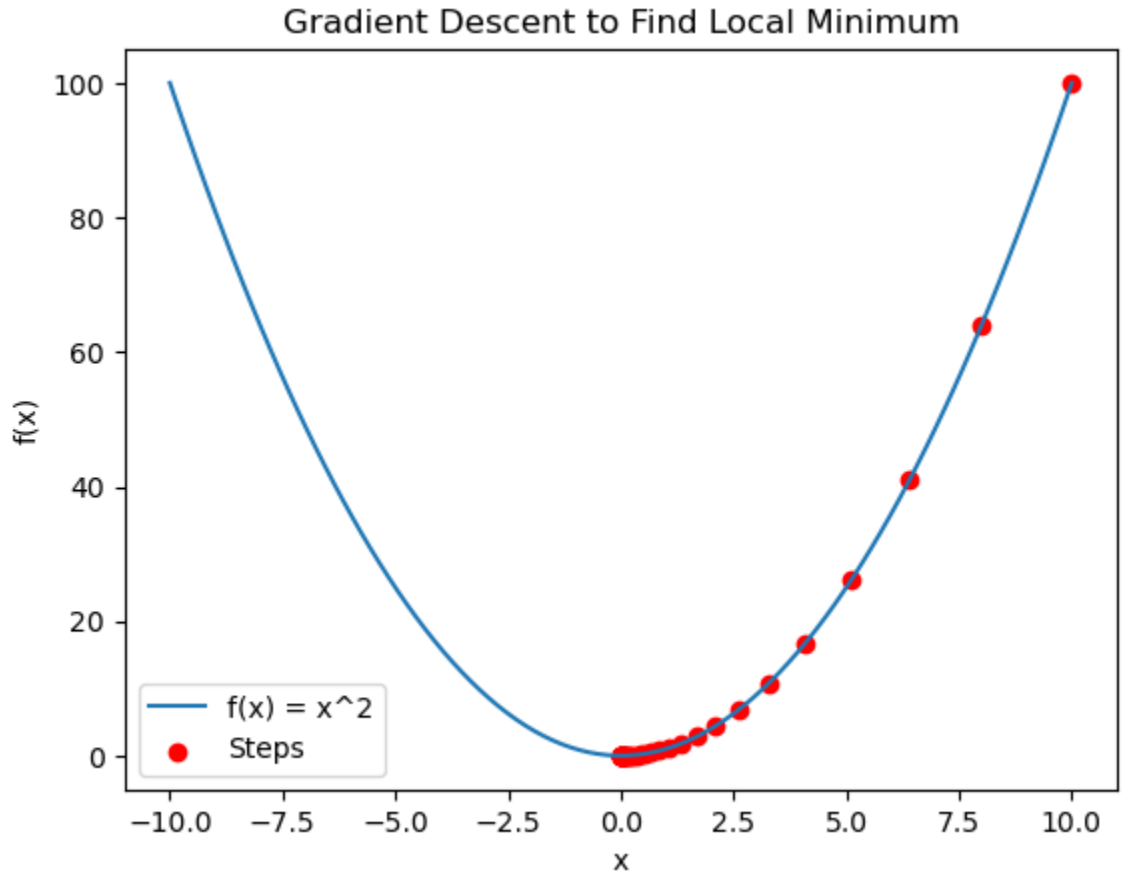
```
x_vals = np.linspace(-10, 10, 100)
y_vals = f(x_vals)
```

In [20]:

```
plt.plot(x_vals, y_vals, label="f(x) = x^2")
plt.scatter(history, [f(x) for x in history], color="red", marker="o", label="Steps")
plt.title("Gradient Descent to Find Local Minimum")
plt.xlabel("x")
plt.ylabel("f(x)")
```

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```
plt.legend()
plt.show()
```



In [ ]:

```

```

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